

Geocomputation & Machine Learning course

Quick recap

This was a GDAL (Geospatial Data Abstraction Library) training session led by Giuseppe, covering advanced raster and vector operations for geospatial data processing. The class focused on virtual raster files (VRT), pixel location extraction using GDAL's new syntax, and multi-core processing techniques. Giuseppe demonstrated how VRT files can significantly reduce processing time and storage requirements by serving as pointers to multiple raster files rather than creating large combined TIFF files. The session covered practical applications including geolocation information extraction, raster resampling, tile aggregation and disaggregation, and vector operations using OGR commands. Students shared their specific use cases, with Abdul working on HEC-RAS simulation results and flood mapping comparisons, Areeba planning supervised classification for land use/land cover analysis using Landsat and Sentinel data, and Eva working on climate and land cover data for fire prediction models. The class also covered multi-core processing techniques using xargs to efficiently handle batch operations and optimize computational resources.

Next steps

Abdul

- Test the multi-core processing approach using xargs with GDAL commands (e.g., GDAL warp, GDAL translate) for his HEC-RAS simulation comparison workflow.
- Crop the large reference flood map from the authority to match the extent of his simulation TIFFs using GDAL Translate, then extract and compare pixel values for validation.

Ali

- Define the master raster file and align other layers to it using GDAL Translate/Warp, selecting appropriate resampling methods according to data type.

Areeba

- Download a few Landsat and Sentinel images, align them (projection/resampling), and prepare them for supervised classification, ensuring temporal alignment.

Eva.sevillano

- Re-check the GDAL edit.py command syntax for changing no-data values and, if still not working, arrange a separate session with Giuseppe to troubleshoot.
- Decide on the target variable (fire/no-fire classification or regression on severity/intensity) for her analysis using climatic and land cover data.

Collaboration

- All participants: Watch the video recording in the syllabus about GDAL and PK tools multi-core operations and try to apply the concepts to their own data processing workflows.
- All participants: Adjust the virtual machine settings to increase the number of allocated CPUs before running multi-core processing tests.

Summary

GDAL Operations and VRT Files

Giuseppe led a discussion on GDAL operations, addressing questions from Abdul and Eva about coding exercises and file handling. Abdul successfully modified a code to create a coordinate table and asked about the "for file in" syntax, which Giuseppe explained as a typical for loop used for processing multiple files. Giuseppe also introduced the concept of VRT (Virtual Raster) files, explaining their use in creating small, pointer-based files that can reference larger TIFF files, and demonstrated how VRTs can be used for mosaicking and band combination. No specific decisions or action items were mentioned, but the session focused on clarifying concepts and addressing technical issues.

Virtual Raster Tables Implementation

Giuseppe explained the use of Virtual Raster Tables (VRTs) in processing large datasets, highlighting their advantages in saving time and space by serving as pointers to operations rather than creating large TIF files. He demonstrated how VRTs can be used to handle independent tiles, recombine them, and perform operations like cropping, resampling, and changing pixel values and no-data levels. Giuseppe also discussed the potential for multi-core processing by splitting operations across tiles and aggregating results back, though he noted that some operations still require manual splitting due to limited built-in multi-core support in GDAL.

GDAL Data Extraction Demonstration

Giuseppe demonstrated how to use GDAL to efficiently extract data from TIFF files using a text file with X and Y coordinates as input. He explained that the GDAL approach is much faster than using while loops because it processes the entire file in one operation rather than looping through each line individually. Giuseppe also highlighted the importance of ensuring coordinates fall within the raster area, noting that points outside the raster area will result in empty pixels, and points in nodata areas will return nodata values.

GDAL Raster Pixel Extraction Methods

Giuseppe demonstrated different methods for extracting pixel values from raster images using GDAL tools, focusing on geolocation info and the new JIRA raster pixel info syntax. He explained that geolocation info is faster for large datasets, while the new syntax requires specifying projection information and can be used with VRT stacks to process multiple files simultaneously. Giuseppe noted that the new method is faster because it opens files only once rather than repeatedly, and mentioned that PKExtract will be covered in future sessions as another complementary tool for extracting raster values.

Data Sources and ML Applications

The team discussed data sources being used for their projects, with Sun mentioning land type data from NLCD and population density, Eva using climatic data and land cover for fire products, and Abdullah working on hydrology modeling. Giuseppe emphasized the importance of defining a clear target variable for machine learning applications, particularly for supervised classification, and explained how to convert images to text files using GDAL for debugging and

analysis purposes. The discussion also covered useful GIS commands like GDAL tile index for checking missing tiles and creating buffers around specific points.

GDAL Geospatial Operations Demonstration

Giuseppe demonstrated GDAL commands for creating proximity and performing geospatial operations, including image compression and data preparation. Areeba inquired about supervised classification for LULC using Landsat and Sentinel images, and Giuseppe advised that significant data preparation work, including image alignment and projection conversion, would be needed before machine learning classification could be applied. Abdul sought guidance on comparing HEC-RAS simulation results with Italian government reference data, and Giuseppe recommended using GDAL Translate for cropping and resampling to match the smaller simulation areas with the larger authority map for correlation analysis.

GIS Image Resolution Techniques

Giuseppe discussed techniques for resolving image resolution issues in GIS analysis, emphasizing the importance of selecting a master file and using appropriate resampling methods based on the data type. He explained that different resampling techniques like nearest neighbor, bilinear, and cubic spline should be chosen depending on the data characteristics, such as temperature surfaces versus digital elevation models. Giuseppe also addressed combining data from different sources like Sentinel-2 and Landsat, noting that timing consistency is more important than resolution matching, and warned about the technical challenges involved in merging images from different sources.

Raster vs Vector Data Processing

Giuseppe discussed the advantages of using raster files over vector files for processing, explaining that raster data is faster to read due to its pixel-based structure. He demonstrated how to use OGR and GDAL tools to work with vector data, including querying, cropping, and converting between different formats. Giuseppe showed examples of using command-line tools to manipulate vector data, such as extracting country information and creating scripts for automatic processing of multiple countries.

GDAL Vector Operations Demonstration

Giuseppe demonstrated how to use OGR and GDAL vector operations for spatial analysis, including filtering, clipping, and buffer operations. Abdul shared his experience using SQL to store CSV data and perform analysis in Spyder with OGR, while Sun discussed using shapefiles to calculate watershed statistics. The discussion covered the advantages of using GDAL vector operations directly in Bash for faster processing compared to Python or R interfaces.

R for Multi-core Processing Techniques

Giuseppe taught the class about using R for computational tasks and emphasized the importance of providing file information when using AI tools like EI for image processing. He demonstrated how to transform a single-core for loop into a multi-core loop using XARG in Bash, showing the efficiency gains when processing multiple files simultaneously across multiple CPU cores. The class was assigned to review the syllabus video recording on GDAL and PK tools, and students were encouraged to test the multi-core processing techniques with their own cases.